

Capital Structure Theories

Theories

- Net Operating Income Approach
- Net Income Approach
- Modigliani Miller Model
- Trade-off Theory
 - Static Trade-off Theory
 - Dynamic Trade-off Theory
- Pecking Order Theory

Net Operating Income Approach

Assumptions

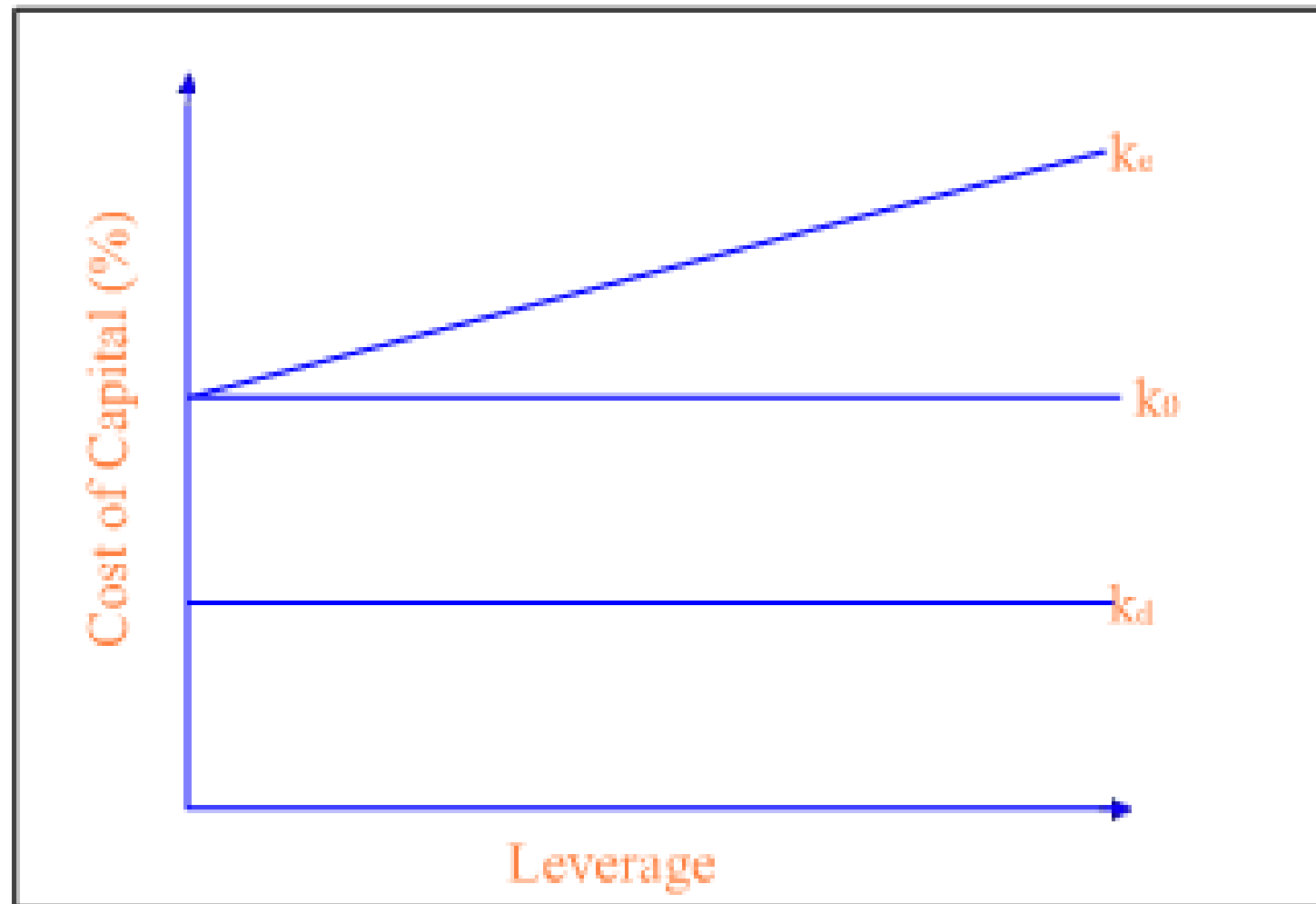
- Market capitalizes the value of the firm as a whole. Thus the split between debt and equity is not important.
- Market uses an overall capitalization rate K_o to capitalize the net operating income. K_o is a constant.
- Debt capitalization rate K_d is a constant.
- Advantage of debt is offset exactly by the increase in the equity capitalization rate, K_e .
- Corporate income taxes do not exist.

Net Operating Income Approach

Implications

- Since the overall cost of capital and cost of debt are constant with increase in debt, the expectations of the equity shareholders raises causing an increase in the cost of equity relative to the increase in proportion of debt in the firm's capital structure.
- Higher debt increases interest obligations of the firm and hence the financial risk is also higher.
- Equity shareholders to compensate for the high financial risk demand higher cost leading to a proportionate rise in the cost of equity as leverage in the capital structure increases.

The effect of leverage on the cost of Capital (NOI approach)



Example

- Assume the firm is investing INR 100,000 to earn a net operating income of INR 60,000. If the firm is considering borrowing debt at 10%, the various components of cost of capital to the firm under the following situations:
- Situation (i) Borrow INR 20,000; Equity contribution INR 80,000
- Situation (ii) Borrow INR 50,000; Equity contribution INR 50,000
- Situation (iii) Borrow INR 80,000; Equity contribution INR 20,000

Cost of Equity: NOI Approach

- **Computation:** $K_o = K_d (D/V) + K_e (E/V)$;
- Situation (i) $60\% = 10\% (.2) + K_e(.8)$; $K_e = 72.5\%$
- Situation (ii) $60\% = 10\% (.5) + K_e(.5)$; $K_e = 110\%$
- Situation (iii) $60\% = 10\% (.8) + K_e(.2)$; $K_e = 260\%$

	Situation (i)	Situation (ii)	Situation (iii)
Cost of capital (K_o)	60%	60%	60%
Cost of debt (K_d)	10%	10%	10%
Cost of equity (K_e)	72.5%	110%	260%

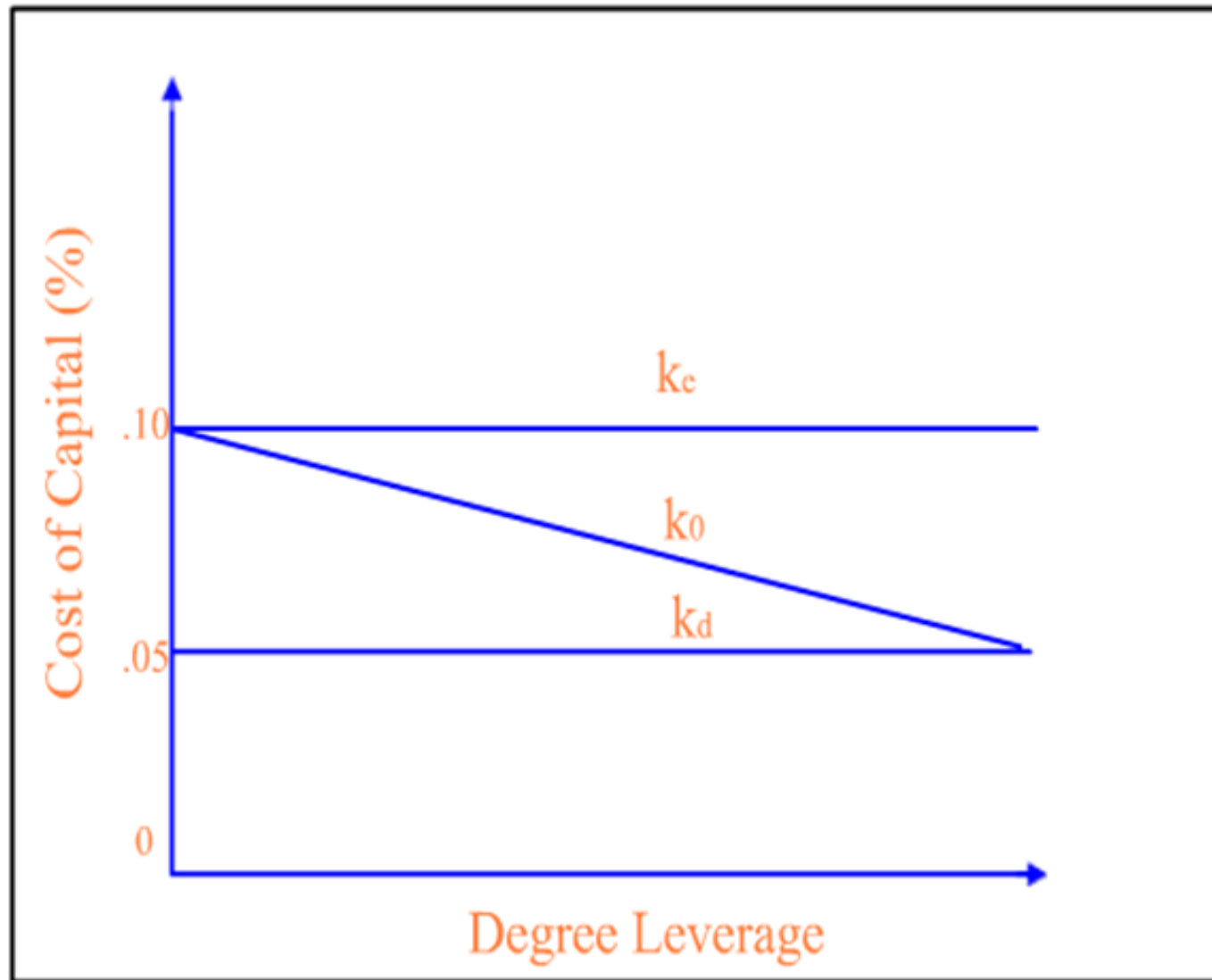
Net Income Approach Assumptions

- Use of debt does not change the risk perception of investors; as a result the equity capitalisation rate K_e and the debt-capitalisation rate, K_d remain constant with change in leverage.
- Debt rate is less than the equity capitalisation rate (i.e., $k_d < k_e$)
- Corporate income taxes do not exist

Net Income Approach Implications

- Net income to the equity shareholders are assumed to be constant. Besides the cost of debt is also assumed to be constant.
- This leads to the hypothesis that as low cost debt component in the capital structure increases, the overall cost of capital will decline.
- Since the overall cost of capital is the weighted average cost of debt and equity, the effect of leverage tends to decrease the overall cost of capital and in turn the best capital structure for the firm will be full debt financing.
- Debt cost being lower, the preference should be only for debt in the capital structure of the firm.

The effect of leverage on the cost of Capital (NI approach)



Example

- Assume a firm invests INR 100,000 and earns a constant net income of 10% to its equity shareholders. Further let us assume that any amount of debt is available to the firm at the cost of 5%. Overall cost of capital under the following situations:
- Situation (i) Borrow INR 20,000; Equity contribution INR 80,000
- Situation (ii) Borrow INR 50,000; Equity contribution INR 50,000
- Situation (iii) Borrow INR 80,000; Equity contribution INR 20,000

Cost of Capital: NI Approach

- **Computation:** $K_o = K_d (D/V) + K_e (E/V)$;
- Situation (i) $K_o = 5\% (.2) + 10\%(.8)$; $K_o = 9\%$
- Situation (ii) $K_o = 5\% (.5) + 10\%(.5)$; $K_o = 7.5\%$
- Situation (iii) $K_o = 5\% (.8) + 10\%(.2)$; $K_o = 6\%$

	Situation (i)	Situation (ii)	Situation (iii)
Cost of capital (K_o)	9%	7.5%	6%
Cost of debt (K_d)	5%	5%	5%
Cost of equity (K_e)	10%	10%	10%

Traditional View Assumptions

- Cost of debt K_d is constant only up to a certain point of debt. Beyond that cost of debt has the tendency to increase proportionately.
- Cost of equity rises with increase in leverage but at a lesser rate.
- Cost of debt is lesser than the cost of equity

Traditional View

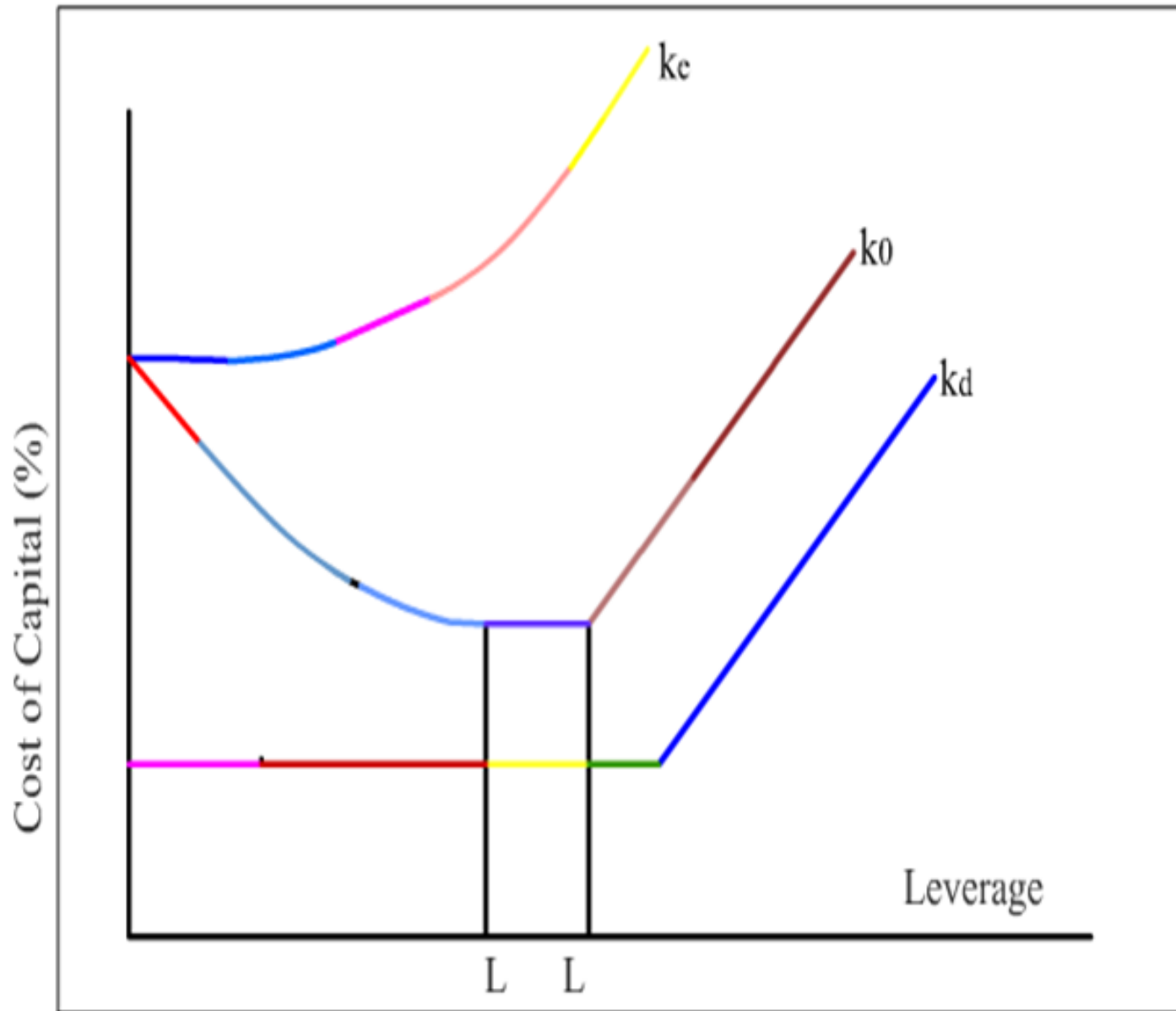
Capital Structure Stages

- Assumptions lead to three stages for a firm in its capital structure change.
- **Stage I** is the initial duration wherein debt is constant: Constant low cost of debt does lead to a decline in the overall cost of capital.
- **Stage II** is where the increase in cost of equity overrides the advantages of low cost of low debt. Firm does not face any more decline in its total cost of capital and hence this leads to a constant total cost of capital for the firm.
- **Stage III** indicates the situation where excess debt leads to excess cost of debt. Proportionate increase in debt as well as equity leads to a rising overall cost of capital.

Traditional View Implications

- Every firm hence has a capital structure which leads to a least cost.
- Optimal capital structure for the firm when the cost of capital is the lowest with changes in debt proportion in the capital structure
- Firm maintains the optimal capital structure for all its projects / divisions until economic scenario changes.

Effect of Leverage on Cost of Capital (Traditional View)



Example

- Assume a firm invests INR 100,000 and earns a constant net income of 15% to its equity shareholders. Further let us assume that any amount of debt is available to the firm at the cost of 5% for 40,000 and 2% for any additional 10,000 debt. Equity cost at 10% and grows at the rate of 2% for every additional 10,000 debt beyond 40,000. Overall cost of capital under the following situations:
- Situation (i) Borrow INR 20,000; Equity contribution INR 80,000
- Situation (ii) Borrow INR 50,000; Equity contribution INR 50,000
- Situation (iii) Borrow INR 80,000; Equity contribution INR 20,000

Cost of Capital: Traditional Approach

- **Computation:** $K_o = K_d (D/V) + K_e (E/V)$;
- Situation (i) $K_o = 5\% (.2) + 10\%(.8)$; $K_o = 9\%$
- Situation (ii) $K_o = 5.4\% (.5) + 12\%(.5)$; $K_o = 8.7\%$
- Situation (iii) $K_o = 9\% (.8) + 18\%(.2)$; $K_o = 10.8\%$

	Situation (i)	Situation (ii)	Situation (iii)
Cost of capital (K_o)	9%	8.7%	10.8%
Cost of debt (K_d)	5%	5.4%	9%
Cost of equity (K_e)	10%	12%	18%

THE MODIGLIANI-MILLER HYPOTHESES

ASSUMPTION:

Perfect capital markets Securities (Shares and debt instruments) are traded in a perfect capital market situation.

- (a) Investors are free to buy or sell securities;
- (b) Investors can borrow without restriction at the same terms as the firms do;
- (c) Transaction costs, (cost of buying and selling securities) do not exist;
- (d) Investors behave rationally.

M-M Hypothesis

- ***Risk:*** Risk of investors is defined in terms of the variability of the net operating income (NOI). Risk of investors depends on both random fluctuations of the expected NOI and the possibility that the actual value may turn out to be different than the best estimate.
- ***Homogeneous risk classes:*** Firms can be grouped into homogeneous risk classes. Firms would be considered to belong to a homogeneous risk class if their expected earnings have identical risk characteristics.

M-M Hypothesis

- ***No taxes*** In the original formation of their hypothesis, M-M assume that no corporate income taxes exist.
- ***Full Payment*** Firms distribute all net earnings to the shareholders, which means a 100 per cent dividend payment.

M-M Proposition I

- For companies in the same risk class, the total market value is independent of the debt-equity mix and is given by capitalizing the expected net operating income by the rate appropriate to that risk class.
- Proposition I concludes Financial Leverage has **no effect** on shareholders' wealth.

Firm Value

- Equity and Debt Values

$$\begin{aligned}\text{Value of the firm} &= \text{Market value of equity} + \text{Market Value of debt} \\ &= \frac{\text{Expected net operating income}}{\text{Expected overall capitalisation rate}}\end{aligned}$$

$$v = (E + D) = \frac{\bar{X}}{K_0} = \frac{NOI}{K_0}$$

Where

V = The market value of the firm

E = The market value of the firm's ordinary equity

D = The market value of debt

$\bar{X}$ = The expected net operating income on the assets of the firm

k_0 = The capitalisation rate appropriate to the risk class of the firm

M-M Proposition II

- M-M's proposition II which defines the cost of equity, follows from their Proposition I.
- Cost of equity formula can be derived from M-M's definition of the average cost of capital.
- **Proposition II** concludes that the rate of return that investors expect to receive on their shares increases as the company's debt-equity ratio increases.

Cost of Equity

- Expected yield on equity or the cost of equity

$$K_e = \frac{\bar{X} - k_d D}{E} \text{ Equation (1)}$$

since

$$K_o = \frac{\bar{X}}{V} \text{ Equation (2)}$$

and K_o and V are constant by definition,

$$\bar{X} = K_o V = K_o (E + D) \text{ Equation (3)}$$

Substituting Equation (3) into Equation (1)

$$K_e = \frac{K_o (E + D) - k_d D}{E} = \frac{K_o E + K_o D - k_d D}{E} = K_o + (K_o - K_d) \frac{D}{E}$$

Firm's Risk

- Increase in expected return is exactly offset by an increase in risk and therefore the shareholders' required rate of return.
- Similar to the weighted cost of capital component, the weighted risk component with respect to beta can be computed as:
- $$\text{Beta of assets} = (\text{proportion of debt} \times \text{beta of debt}) + (\text{proportion of equity} \times \text{beta of equity})$$

Risk of a Levered Firm

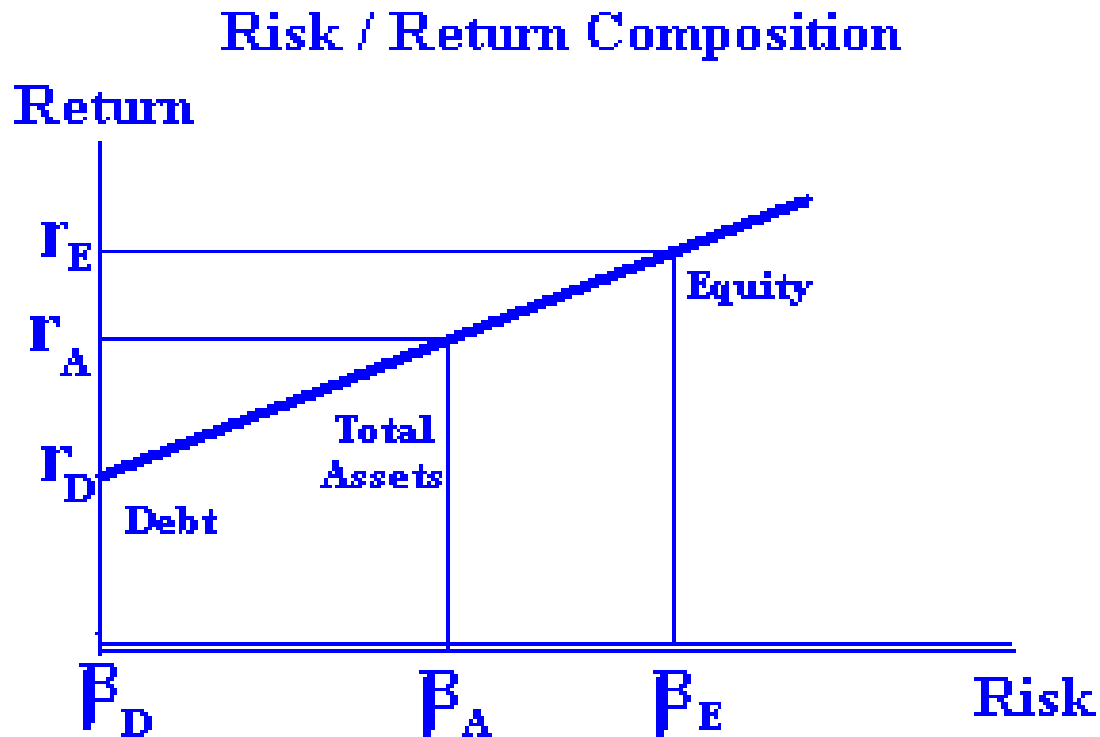
$$\beta_A = \left(\frac{D}{D+E} \times \beta_D \right) + \left(\frac{E}{D+E} \times \beta_E \right)$$

- Rearrange this equation to give an expression of **beta of equity of a levered company**

$$\beta_E = \beta_A + \frac{D}{E} \times (\beta_A - \beta_D)$$

Asset Risk Vs Equity Risk

- Debt increases (numerator), the risk of equity holders also increases.



M-M Hypothesis

Supportive Arguments

- M-M supported financial irrelevance through the following arguments:
- ***Arbitrage Process***
- ***Home-made Leverage***
- ***Firm's Total Returns***

Arbitrage Process

- Assuming a company which has financed its investment requirement through debt is of more value to investors

GIVEN THE CAPITAL MARKET ASSUMPTIONS,

- Any rational investor would try to switch over from the existing all equity company to the more valued debt-structured company.
- This will increase demand for the share of debt-structured company shooting up the market price of the share.
- Arbitrage process results in investors paying more in the secondary market (in terms of competitive pricing in the market) to gain the advantage of debt, while the all equity investors pay less in the secondary market (in terms of competitive pricing between investors) to compensate for the lesser valuation for not having debt in their capital structure.
- Arbitrage process will stop when there is no opportunity for the investors to gain from the difference in valuation.

Home-made Leverage

- M-M also introduced the concept of investors opting for same percentage of investment in their capital structure as that of the company whether levered or unlevered.
- Through this, the investor gains the same advantage of holding debt similar to that of the firm which holds debt in its capital structure.
- Personal gain compensates the gain achieved by the firm through introduction of debt.

Firm's Total Returns

- Investor has the option of investing in either debt or equity in the firm. If the investor invests 1% of the total investments in unlevered firm, (investment and profits same), the returns paid to the investor and the contribution in terms of investment is [0.01 : 0.01]

UNLEVERED

Equity	Profit
0.01	0.01

- If the firm is levered, investor has the option of participating in both the debt and equity component of the firm in the same proportion and hence, the investor's investment contribution and return will appear as [0.01 : 0.01]

LEVERED

	Value	Return	
Debt	0.005	0.005	Interest
Equity	0.005	0.005	Profit
Total	0.01	0.01	

M-M Proposition II

- Return to investor in a levered firm is the same as that of the investor who invested in an all equity firm.
- Capital structure decisions are **irrelevant** to the financial manager of a company.

Implications of Tax on M-M Theory

- Value of firm L is equal to the value of the unlevered firm plus the present value of the interest tax shield.
- Value of the unlevered firm is the value of total assets.
- Total assets are represented by total equity capital.
- Value of levered firm = Value of unlevered firm + PV of tax shield

Value of Levered Firm

- When the corporate tax rate, T , is positive ($T > 0$) the value of the levered firm will increase continuously with debt.
- $V_t = V_a + TD$
- $V_t = V_u + TD$
- V_t : Value of firm with debt
- T : Corporate tax rate
- D : Debt Interest

Value of Levered Firm

- Theoretically the value of the firm will be maximum when it employs 100 per cent debt.
- $$V_t = \frac{\bar{X}(1-T)}{K_u} + \frac{TK_d D}{K_d}$$
- $\bar{X}$: Perpetual operating income stream of all equity firm
- K_u : Capitalization rate of all equity firm
- K_d : Cost of debt
- D : Debt value

Example

	Unlevered Firm	Levered Firm
Net Operating Income Before Tax	150,000	150,000
Less Interest (700,000 @ 7%)		49,000
Taxable Income	150,000	101,000
Tax @ 35%	52,500	35,350
Income after tax	97,500	65,650`
Add: Interest (Payment to debt holders)		49,000
Total Income payment to equity & debt holders	97,500	114,650
Cost of capital	14%	
Firm Value	696,429	696,428
Tax Shield (perpetual)		245,000*
Total Firm Value	696,429	941,428

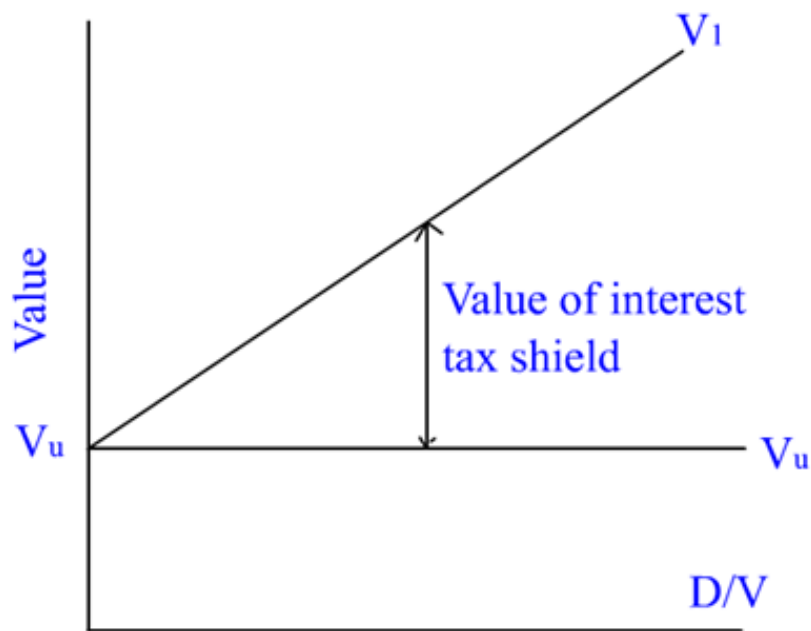
Perpetual tax shield = Tax rate of Interest paid discounted at interest rate = (Debt interest x tax rate) / 0.07 = (19000 x 0.35) / 0.07

Cost of Levered Firm

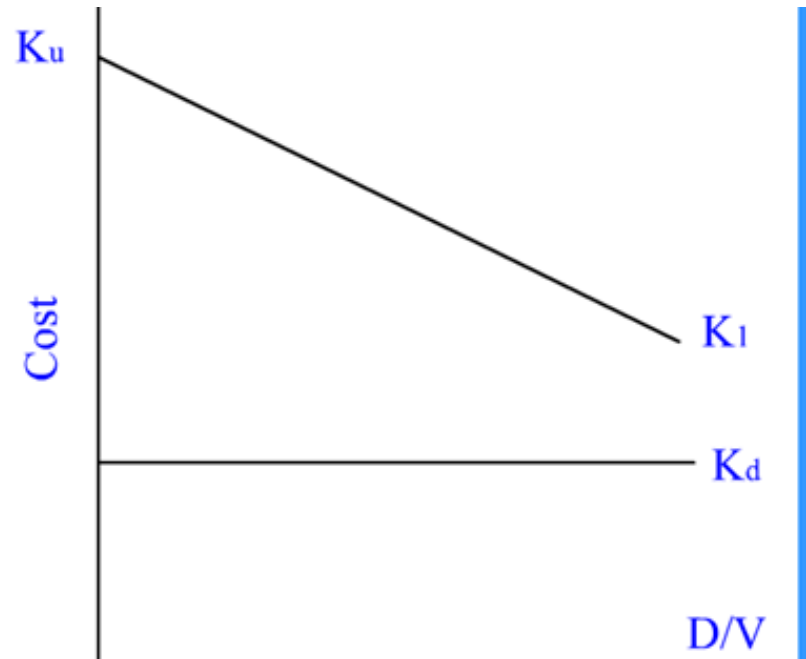
- Under the assumption of the M-M hypothesis with corporate taxes, the levered firm's cost of capital is proportional to debt.
- $K_t = K_u(1 - TL)$
- Where K_t is the levered firm's cost of capital, K_u is the all equity capitalisation rate, T is the corporate tax rate and L is debt ratio.

M-M theory with Taxes

Value of Levered Firm



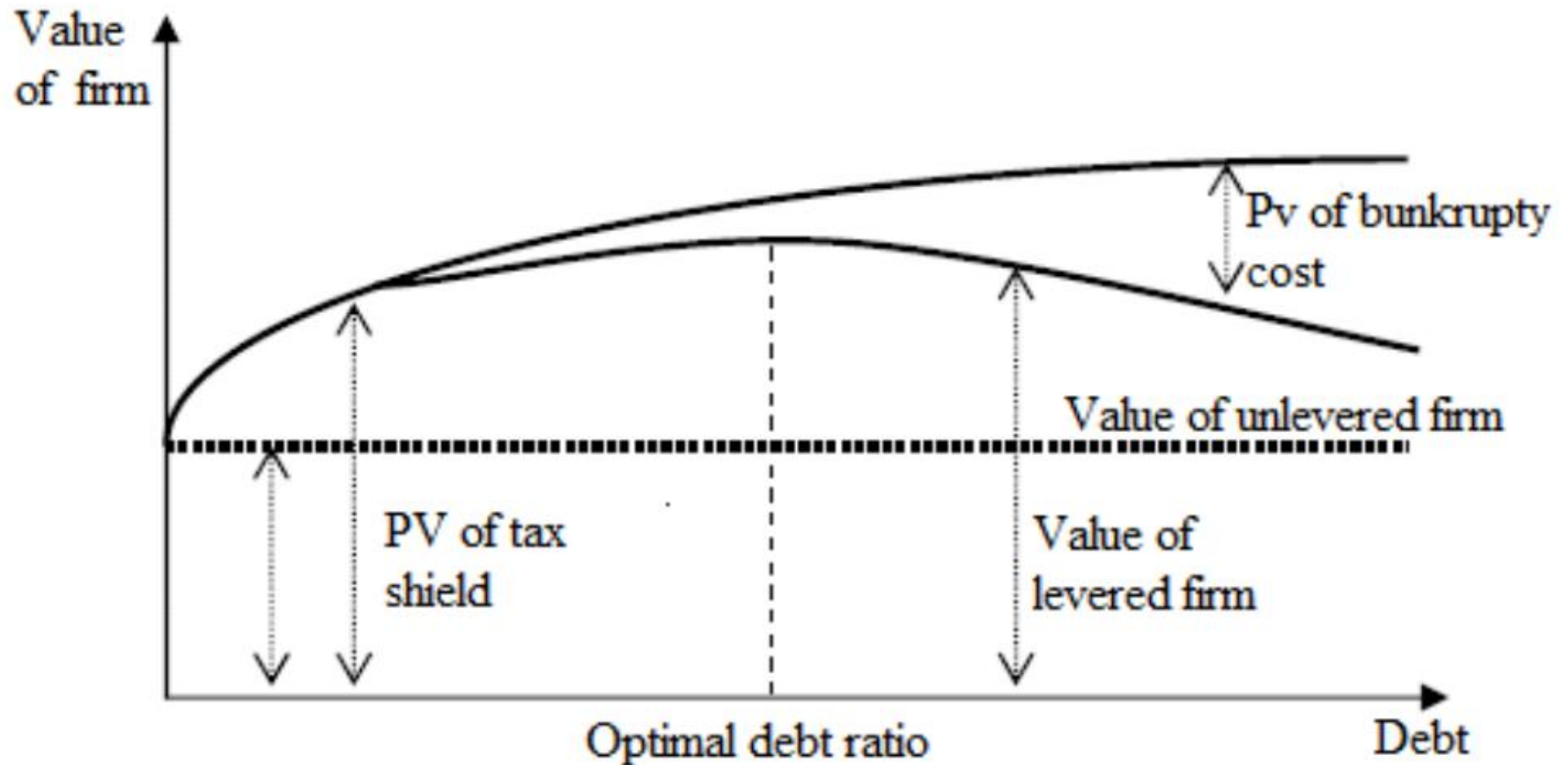
Cost of Levered Firm



Trade-off Theory

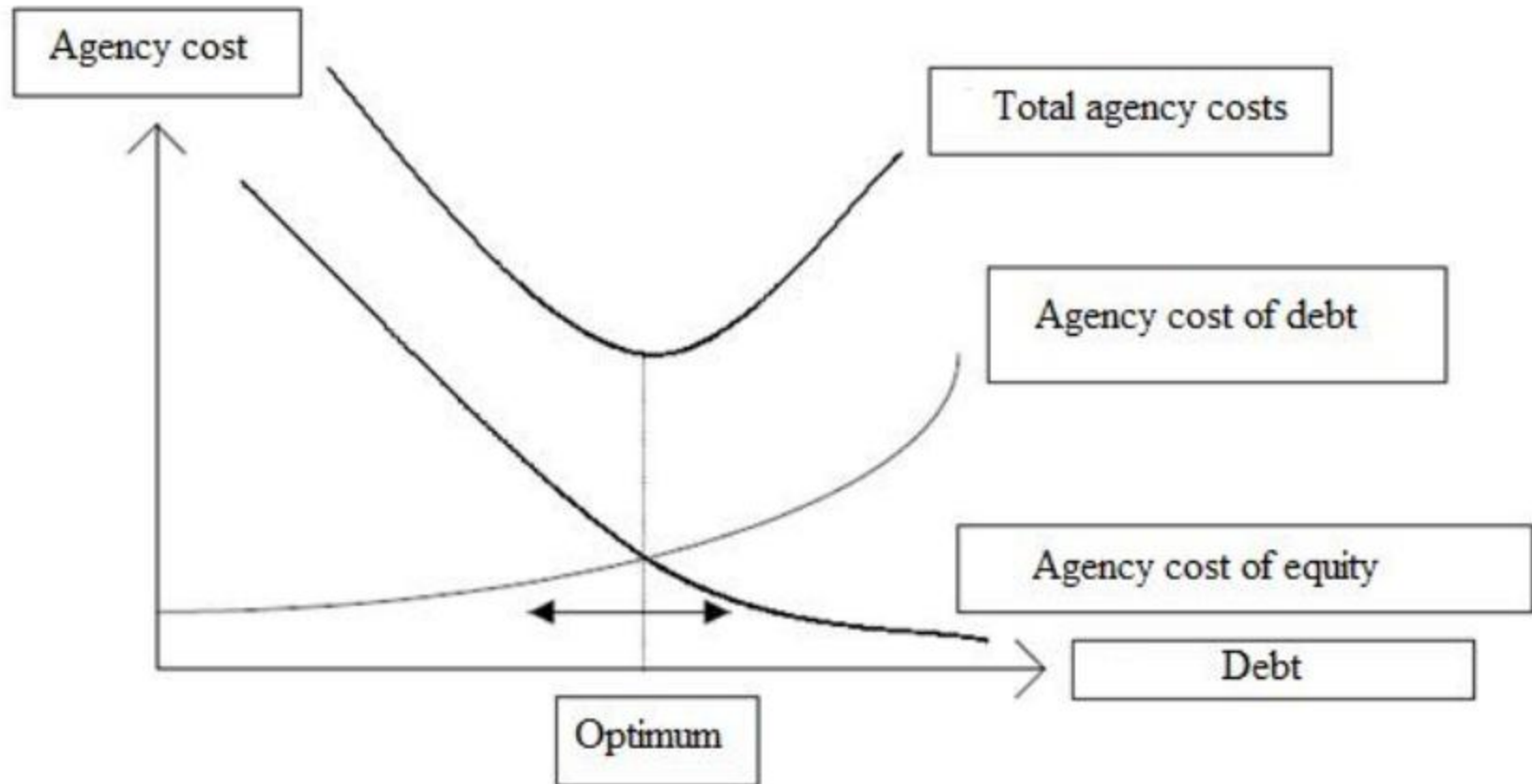
- Optimal capital structure is a trade-off between interest tax shields and cost of financial distress
- $\text{Value of firm} = \text{Value if all-equity financed} + \text{PV}(\text{tax shield}) - \text{PV}(\text{cost of financial distress})$
- Starting point is the value of the all-equity financed firm.
- Present value of tax shields is then added.
- $\text{PV}(\text{tax shield})$ initially increases as the firm borrows, until additional borrowing increases the probability of financial distress rapidly.
- Cost of financial distress is assumed to increase with the debt level.

Static Trade off Theory



Source: Myers S.C., (1984), The capital structure puzzle, The Journal of Finance, Vol.39(3), 577

Agency Cost of Debt



Source: Jensen M.C., (1976), Theory of the firm: Managerial behavior, Agency cost and ownership structure, Journal of Financial Economics, Vol.3(4), 355.

Trade off Theory Evaluation

- Capital structure is based on a trade-off between tax savings and distress costs of debt.
- Firms with safe, tangible assets and plenty of taxable income to shield should have high target debt ratios.
- Theory is capable of explaining why capital structures differ between industries, whereas it cannot explain why profitable companies within the industry have lower debt ratios
- (Trade-off theory predicts the opposite as profitable firms have a larger scope for tax shields and therefore subsequently should have higher debt levels)

Dynamic Trade-off Theory

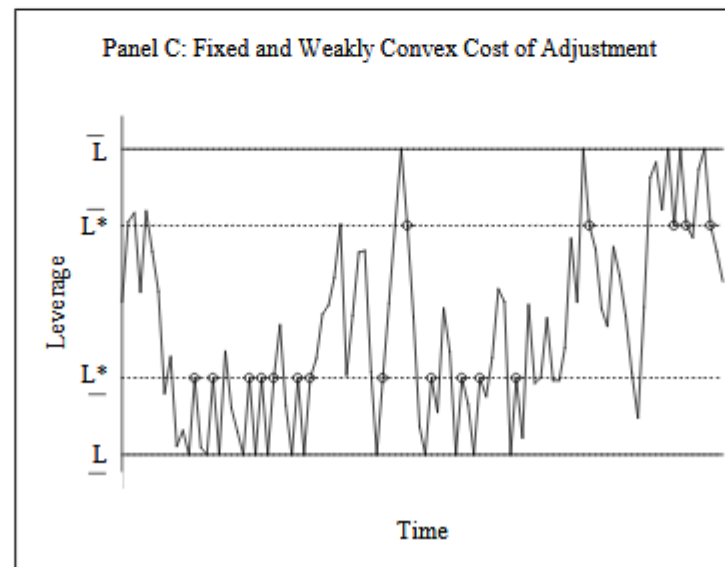
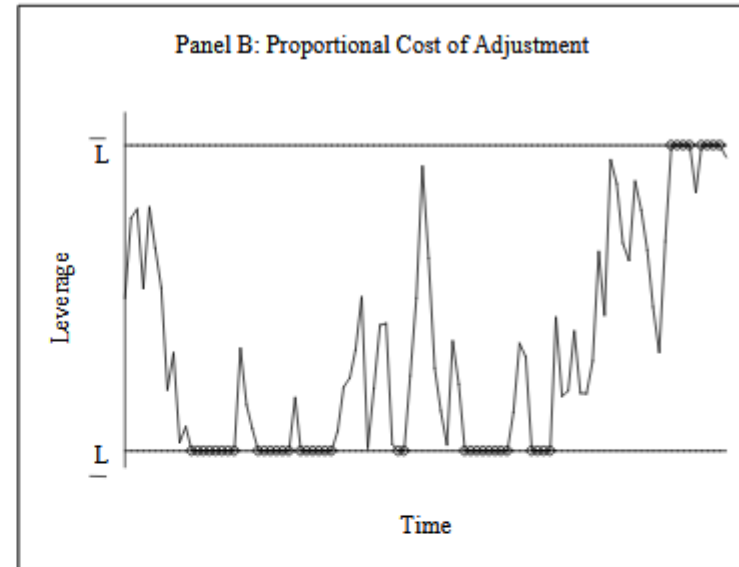
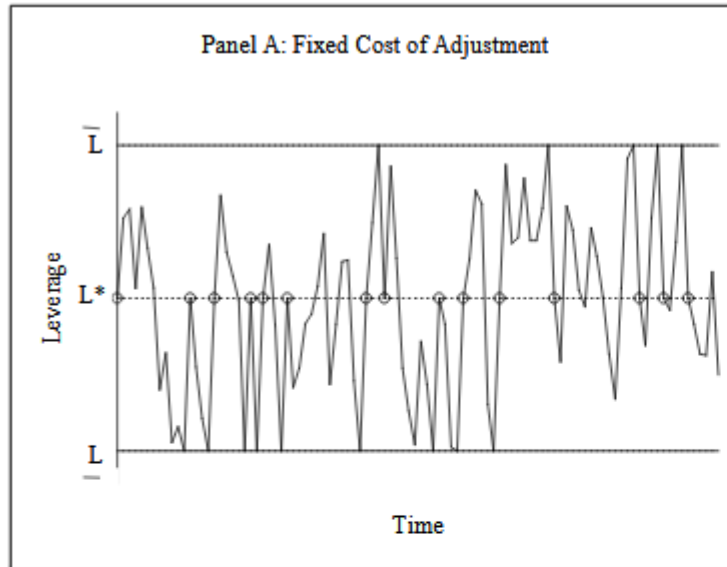
- Leverage Dynamics
 - Impact of adjustment costs
 - Dynamic Trade-off predictions
- Costs and benefits of debt

Source: Leary M.T., and Roberts M.R., (2005), Do firms rebalance their capital structure? The Journal of Finance, 60, 2575-2519

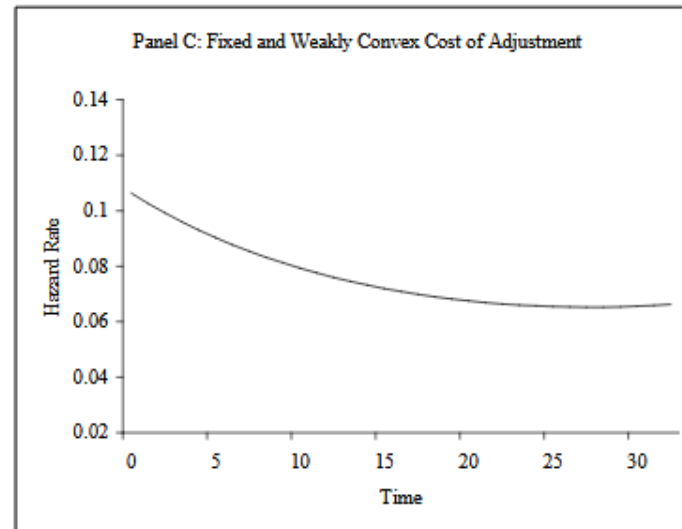
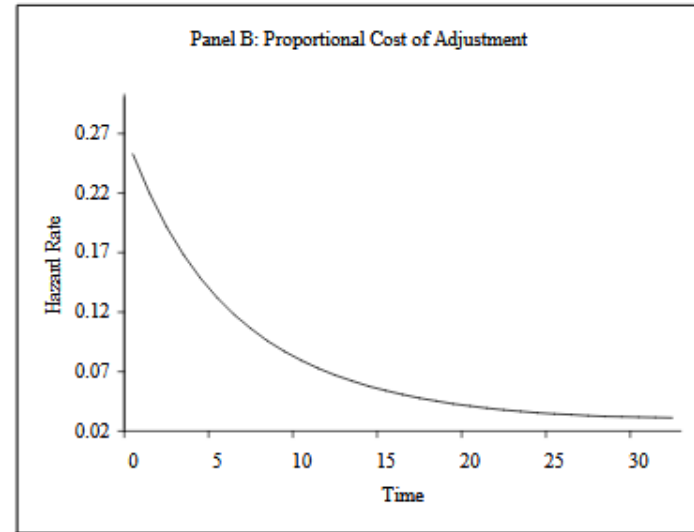
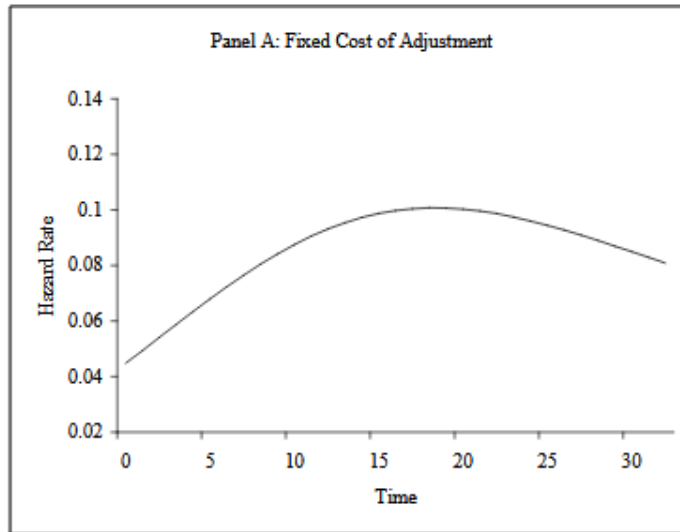
Dynamic Trade-off Theory

- Corporation's capital structure changes can reflect different economic conditions.
- Trade-Off Theory is rather dynamic, with firms rebalance their capital structure to differing optimal ratios during differing economic climates.
- A small problem with a firm's cash flow could lead them to default on its debt. Consequently, it is vital to have less debt than if the economy were in a boom, as this will avoid the costs of financial distress.
- High leverage is an important characteristic of financially distressed firms.
- Another reason for this shift away from debt is that major banks have been put under pressure by the government and regulators to hold more capital to counter potential losses, and to undertake debt-for-equity swaps.
- Corporations that are going to banks for loans are being required to have high levels of equity compared to debt, so the bank can rely on equity as collateral.
- Large interest rates are being charged at a level significantly higher than the LIBOR. Consequently, it is getting harder to acquire debt.

Simulation of Dynamic Trade-off



Simulated Hazard Curves



Pecking Order Theory

- Alternative theory to the trade-off theory. Rather than introducing corporate taxes and financial distress into the M-M framework, the key assumption of the pecking order theory is asymmetric information.
- Asymmetric information captures that managers know more than investors and their actions therefore provides a signal to investors about the prospects of the firm.

Pecking Order Theory Arguments

- If the firm announces a stock issue it will drive down the stock price because investors believe managers are more likely to issue when shares are overpriced.
- Firms prefer to issue debt as this will allow the firm to raise funds without sending adverse signals to the stock market.
- Debt issues might create information problems if the probability of default is significant, since a pessimistic manager will issue debt just before bad news get out.

Pecking Order Theory

- Pecking order of financing decision:
 - Internal cash flow
 - Issue debt
 - Issue equity
- Internal financing is preferred over external financing, and if external finance is required, firms should issue debt first and equity as a last resort.
- Pecking order seems to explain why profitable firms have low debt ratios
 - This happens not because they have low target debt ratios, but because they do not need to obtain external financing.
- Unlike the trade-off theory the pecking order theory is capable of explaining differences in capital structures within industries.